

E-COMMERCE ORDER FULFILMENT DATA PLATFORM

CAPSTONE PROJECT REPORT

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Dataset: Brazilian E-Commerce Public Dataset by Olist

Source URL: <https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>

GitHub: <https://github.com/shashims691-ui/Hero-airflow-project>
<https://github.com/shashims691-ui/E-Commerce-Order-Fulfilment-Data-Platform-coding-part-of-capstone-project>

TABLE OF CONTENTS

1. Introduction
2. Business Problem
3. Project Objectives
4. Bronze Layer
5. Silver Layer
6. Gold Layer
7. Data Model
8. Data Quality Validation
9. Airflow Orchestration
10. Dashboard Development
11. Business Insights
12. Data Governance
13. Performance Optimization
14. Conclusion
15. Future Enhancements

INTRODUCTION

The E-Commerce Order Fulfilment Data Platform project aims to build an end-to-end data engineering solution using the Brazilian E-Commerce Dataset by Olist.

The project focuses on collecting raw operational data, performing data cleaning and validation, creating analytical data models, and generating business insights through interactive dashboards.

The solution follows the Bronze-Silver-Gold architecture and demonstrates modern data engineering concepts including data quality validation, dimensional modelling, orchestration, and business intelligence reporting.

Tool	Purpose
Python	ETL Development
Pandas	Data Cleaning
SQLite	Data Storage
Apache Airflow	Workflow Orchestration
Power BI	Dashboard
GitHub	Version Control

BUSINESS PROBLEM

E-commerce companies receive thousands of customer orders every day. The operational data is generated from multiple sources such as customers, products, sellers, payments, reviews, and order transactions.

Since the data exists in different files, it becomes difficult to analyze business performance and make quick decisions.

The major business problems include:

- Difficulty in tracking order fulfilment.
- Monitoring delivery performance.
- Measuring seller performance.
- Understanding customer satisfaction.
- Identifying preferred payment methods.

Therefore, a centralized data platform is required to integrate, clean, transform, and analyze the data to support business decision-making.

PROJECT OBJECTIVES

The objectives of this project are:

- Build an end-to-end Data Engineering pipeline.
- Store raw data in the Bronze Layer.
- Perform data cleaning and validation in the Silver Layer.
- Create analytics-ready Fact and Dimension tables in the Gold Layer.
- Implement data quality validation rules.
- Design an Airflow DAG for ETL orchestration.
- Store processed data for analytical reporting.
- Develop interactive Power BI dashboards.
- Generate business insights using KPIs and visualizations.
- Demonstrate the complete Data Engineering lifecycle from ingestion to visualization.

DATASET DESCRIPTION

The project uses the Brazilian E-Commerce Public Dataset (Olist) obtained from Kaggle.

The dataset contains transactional information about orders, customers, sellers, products, payments, and customer reviews.

For faster development and testing, a three-month subset of the dataset was used during implementation while preserving the complete data engineering workflow.

<i>DATASET NAME</i>	<i>DESCRIPTION</i>
<i>Orders</i>	<i>Contains customer order details</i>
<i>Customers</i>	Customer information and location
<i>Sellers</i>	Seller information
<i>Products</i>	Product details
<i>Order Items</i>	Product-wise order information
<i>Payments</i>	Payment type and payment amount
<i>Reviews</i>	Customer ratings and reviews

Dataset	Records
Orders	99,441
Customers	99,441
Order Items	112,650
Payments	103,886
Reviews	99,224
Products	32,951
Sellers	3,095

BRONZE LAYER

The Bronze Layer is the first stage of the data engineering pipeline.

Purpose:

- Store raw data exactly as received.
- Preserve original source files.
- Maintain historical records.

Activities Performed:

- Downloaded Olist dataset.
- Loaded CSV files into Google Colab.
- Stored raw files without modification.

Input Files:

- Orders
- Customers
- Sellers
- Products
- Payments
- Reviews
- Order Items

No data cleaning or transformation was performed in this layer.

SILVER LAYER

The Silver Layer is responsible for cleaning and transforming the raw data.

Activities Performed:

- Converted timestamp columns into datetime format.
- Removed duplicate records.
- Checked for null values.
- Verified data types.

- Validated payment values.
- Validated freight values.
- Calculated delivery_days.
- Calculated delay_days.

Data Quality Checks:

- Duplicate Order IDs
- Negative Payment Values
- Negative Freight Values
- Negative Product Prices

Result:

The cleaned data became suitable for analytical processing in the Gold Layer.

GOLD LAYER

The Gold Layer contains analytics-ready datasets.

Dimension Tables:

- dim_customer
- dim_seller
- dim_product

Fact Tables:

- fact_orders
- fact_payments
- fact_reviews
- fact_delivery_sla

The Gold Layer follows a dimensional model that improves reporting performance and simplifies business analysis.

These tables were exported and connected to Power BI for dashboard development.

DIMENSIONAL DATA MODEL

The Gold Layer follows a Star Schema data model consisting of Fact Tables and Dimension Tables.

FACT TABLES

Fact tables store measurable business events.

The following fact tables were created:

- fact_orders
- fact_payments
- fact_reviews
- fact_delivery_sla

DESCRIPTION

fact_orders contains order-level information including order status, purchase date, estimated delivery date, delivery days, and delay days.

fact_payments stores payment-related information such as payment type and payment amount.

fact_reviews stores customer review scores.

fact_delivery_sla stores delivery performance metrics.

DIMENSION TABLES

Dimension tables provide descriptive information.

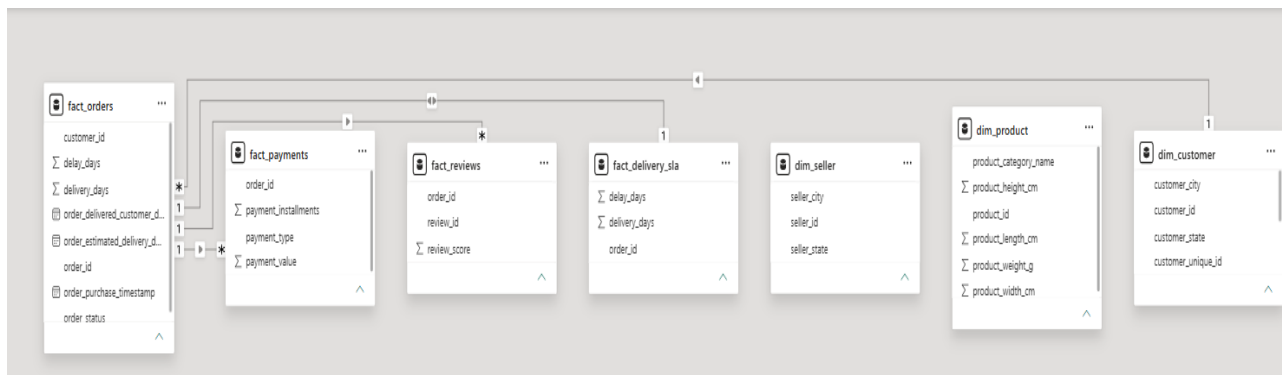
The following dimension tables were created:

- dim_customer
- dim_seller
- dim_product

These dimension tables were connected with fact tables using primary and foreign key relationships in Power BI to support analytical reporting.

The dimensional model improves query performance and simplifies dashboard development.

Star Schema Data Model



DATA QUALITY VALIDATION

Data quality plays an important role in building reliable analytical systems.

Several validation rules were implemented before loading the data into the Gold Layer.

VALIDATION RULES

1. Order IDs must be unique.
2. Product prices must be non-negative.
3. Freight values must be non-negative.
4. Payment values must be non-negative.
5. Purchase date must occur before delivery date.
6. Missing values were identified and handled appropriately.

VALIDATION RESULT

Duplicate Order IDs : 0

Negative Product Prices : 0

Negative Freight Values : 0

Negative Payment Values : 0

Datetime Conversion : Successful

Duplicate Records : None

The validation results confirmed that the processed datasets were suitable for analytical reporting.

Validation	Status
Duplicate Orders	Passed
Negative Prices	Passed
Negative Freight	Passed
Negative Payments	Passed
Datetime Validation	Passed
Null Handling	Completed

AIRFLOW ORCHESTRATION

Apache Airflow was selected as the orchestration tool for automating the ETL pipeline.

The Airflow DAG coordinates the execution of different pipeline stages in the correct sequence.

The workflow consists of the following tasks:

- Raw Data Ingestion
- Data Quality Validation
- Silver Layer Transformation

- Gold Layer Creation
- Export Processed Data
- Load Gold Layer Tables into SQLite Database

The DAG supports scheduling, task dependencies, retries, logging, and failure handling.

Airflow ensures that every pipeline step is executed automatically and reliably without manual intervention.

Welcome

Stats

Failed Days 0 Running Days 0 Active Days 1

First 10 favorite Dags

No favorites yet. Click the star icon next to a Dag in the list to add it to your favorites.

Health

MetaDatabase Scheduler Triggerer Dag Processor Pool Slots 128

History

Last 24 Hours 2026-06-18 22:36:06 - 2026-06-19 22:36:06

Dag Runs

State	Count	Percentage
Queued	0	0.00%
Running	0	0.00%
Success	1	33.33%
Failed	2	66.67%

Asset Events

No Asset Events found.

AIRFLOW HOME PAGE

ecommerce_etl_pipeline

Schedule Latest Run Next Run Max Active Runs Owner Tags Latest Dag Version

2026-06-19 21:49:16 0 of 16 Shashikanth v2

Overview Runs Tasks Calendar Audit Log Code Details

ETL PIPELINE RUN SUCCESSFULLY

+ Filter

Dag Run ID	Run After	State	Run Type	Triggering User	Start Date	Duration	Dag Version(s)
manual_2026-06-19T16:19:16.298653+00:00	2026-06-19 21:49:16	Success	Manual	admin	2026-06-19 21:49:17	00:00:09.312	v2

Dag: ecommerce_etl_pipeline | Dag Run: All Runs | Task: load_gold_tables

DAG RUN SUCCESSFULLY

Search Dags Ctrl+K | Trigger

load_gold_tables

Operator: BashOperator

Trigger Rule: all_success

00:05:08 | 00:02:34

Overview | Task Instances | Audit Log

EXPLORER

- ✓ **HERO-AIRFLOW-PROJECT [COD...**
 - > airflow-venv
 - ✓ dags
 - ecommerce_etl_dag.py
 - ✓ data
 - dim_customer.csv
 - dim_product.csv
 - dim_seller.csv
 - fact_delivery_sla.csv
 - fact_orders.csv
 - fact_payments.csv
 - fact_reviews.csv
 - ✓ database
 - ecommerce.db
 - ✓ etl
 - load_to_sqlite.py
 - ⓘ README.md

DASHBOARD DEVELOPMENT

Power BI was used to build an interactive dashboard for business users.

The dashboard provides a high-level overview of order fulfilment performance through Key Performance Indicators (KPIs) and analytical visualizations.

The dashboard supports filtering by

- Order Status
- Payment Method
- Review Score

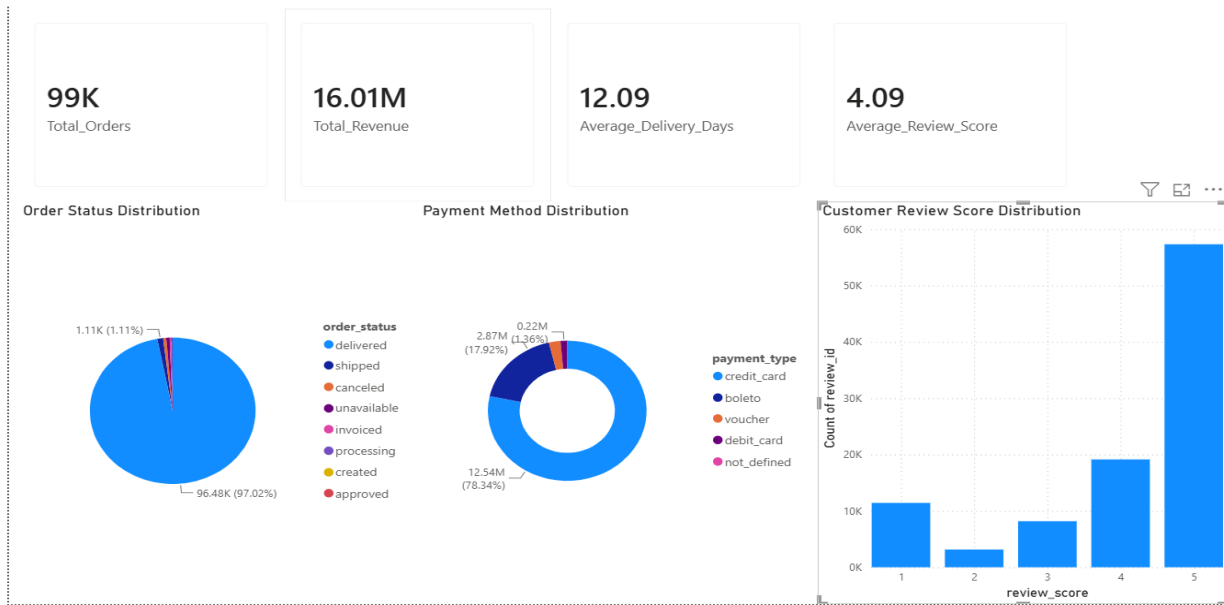
KPI CARDS

- Total Orders
- Total Revenue
- Average Delivery Days
- Average Review Score

DASHBOARD VISUALIZATIONS

- Order Status Distribution
- Payment Method Distribution
- Customer Review Score Distribution

The dashboard enables decision-makers to monitor operational performance, customer satisfaction, and revenue trends in a single interface.



BUSINESS INSIGHTS

The Power BI dashboard generated several valuable business insights from the processed data.

1.ORDER VOLUME

The platform processed approximately 99,000 customer orders. This indicates that the dataset represents a large-scale e-commerce business with significant transaction activity.

2.REVENUE

The total revenue generated from customer payments was approximately 16.01 Million. This KPI helps management understand the overall business performance.

3.DELIVERY PERFORMANCE

The average delivery time was approximately 12.09 days.

Most orders were delivered before or close to the estimated delivery date, indicating an efficient fulfilment process.

4.CUSTOMER SATISFACTION

The average customer review score was 4.09 out of 5.

Most customers provided 5-star ratings, indicating a high level of customer satisfaction.

5.PAYMENT METHODS

Credit Card was the most preferred payment method among customers, followed by Boleto and Voucher payments.

6.ORDER STATEMENTS

The majority of orders were successfully delivered, while only a small percentage were cancelled or unavailable.

These insights help business users identify operational strengths and improvement opportunities.

DATA GOVERNANCE

Data Governance ensures that data remains accurate, secure, consistent, and reliable throughout the pipeline.

Data Dictionary

A data dictionary was prepared to document important business fields.

Column Name	Description
order_id	Unique Order Identifier
customer_id	Unique Customer Identifier
seller_id	Unique Seller Identifier
product_id	Unique Product Identifier
payment_value	Total Payment Amount
freight_value	Shipping Cost
review_score	Customer Rating
delivery_days	Actual Delivery Time
delay_days	Difference between Actual and Estimated Delivery

Data Lineage

Raw CSV Files

↓

Bronze Layer

↓

Silver Layer

↓

Gold Layer

↓

Power BI Dashboard

Access Assumptions

Business Users:

Read-only dashboard access.

Operations Team:

Access to Gold Layer tables.

Data Engineering Team:

Access to Bronze, Silver, and Gold layers.

Personally identifiable information was not exposed in the dashboard.

PERFORMANCE AND COST OPTIMIZATION

Several optimization techniques were implemented to improve pipeline performance.

Dataset Optimization

Instead of processing the complete dataset during development, a three-month subset was used.

Benefits:

- Faster processing
- Reduced memory consumption
- Faster dashboard refresh

- Easier debugging

Data Model Optimization

Fact and Dimension tables were created to reduce query complexity.

This improved reporting performance inside Power BI.

ETL Optimization

Data cleaning and transformation were completed before dashboard development.

This reduced repeated calculations during reporting.

Future Optimization

Future improvements may include:

- Incremental loading
- Partitioning
- Indexing
- Real-time streaming
- Distributed processing using Apache Spark

CONCLUSION

The E-Commerce Order Fulfilment Data Platform successfully demonstrates the complete Data Engineering lifecycle.

The project collected raw operational data, cleaned and validated the data, created analytics-ready fact and dimension tables, and generated meaningful business insights using Power BI dashboards.

The Bronze, Silver, and Gold architecture improved data quality and reporting performance.

The Power BI dashboard enables business users to monitor revenue, delivery performance, customer satisfaction, and order fulfilment efficiently.

Overall, the project provides a scalable foundation for future e-commerce analytics and automation using Apache Airflow.

ARCHITECTURE DIAGRAM



REFERENCES

- Olist Brazilian E-Commerce Dataset

<https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>

- Apache Airflow Documentation

<https://airflow.apache.org/>

- Power BI Documentation

<https://learn.microsoft.com/power-bi/>

- SQLite Documentation

<https://sqlite.org/docs.html>

- Pandas Documentation

<https://pandas.pydata.org/docs/>

THANK YOU